

Chapter 35: Stillbirth — Revision Notes

Williams Obstetrics 26e, pp. 1612–1632 of the PDF. High-yield numbers are in **bold**. The three tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: In the US, fetal deaths ≥ 20 wk now slightly **outnumber neonatal deaths**. Systematic evaluation (**autopsy + placental pathology + genetics by CMA**) finds a probable or possible cause in **$\sim 3\%$** . **Prior stillbirth** is the strongest predictor of recurrence. Deliver a woman with a prior stillbirth at **39 wk**.

1. Perinatal Mortality Overview

- **Perinatal mortality = fetal + neonatal deaths**; a measure of healthcare quality before, during and after delivery.
- US vital statistics usually count fetal deaths **after 20 wk**. In **2013**, fetal deaths slightly outnumbered neonatal deaths.

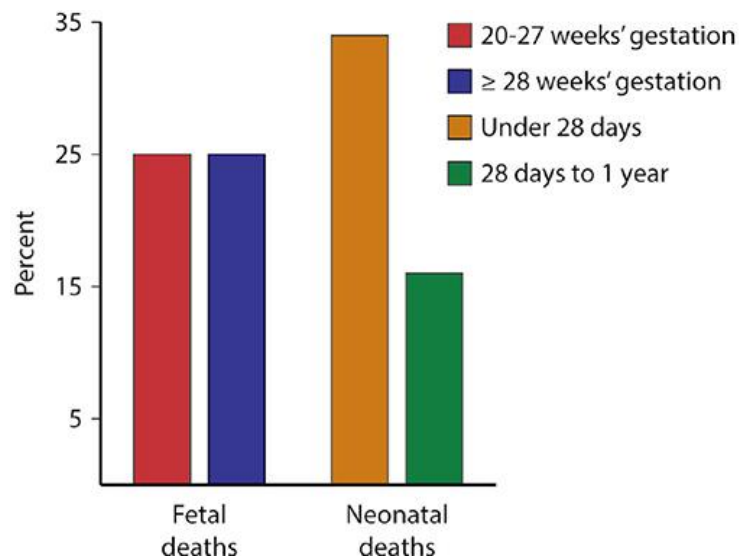


Figure 35-1. US 2013: fetal deaths split ~25% at 20–27 wk and ~25% at ≥ 28 wk; infant deaths ~34% under 28 days and ~16% at 28 days–1 year.

2. Definitions

- **Fetal death (NCHS, based on WHO):** death before complete expulsion or extraction, **irrespective of gestational age**, and **not an induced termination**. After delivery the fetus shows **no breathing, heartbeat, cord pulsation or definite voluntary movement**.
 - Heartbeats $\neq$ transient cardiac contractions; respirations $\neq$ fleeting efforts or gasps.
- **Reporting is set by each state:** most require reporting at **≥ 20 wk** and/or **≥ 350 g** (≈ 20 -wk weight). Under-reporting is common, especially early.
- International comparisons are limited: **$<5\%$** of neonatal deaths worldwide are formally documented.
- Defining US stillbirth by **≥ 500 g** instead of 20 wk would lower the rate by **40%**.

Fetal mortality epoch	Gestational age
Early	<20 completed weeks
Intermediate	20–27 wk
Late	≥28 wk

- Rates in each epoch have changed little since **2006**.
- **Fetal death rates are highest at the earliest and latest GAs** (U-shaped curve) → suggests different causes.

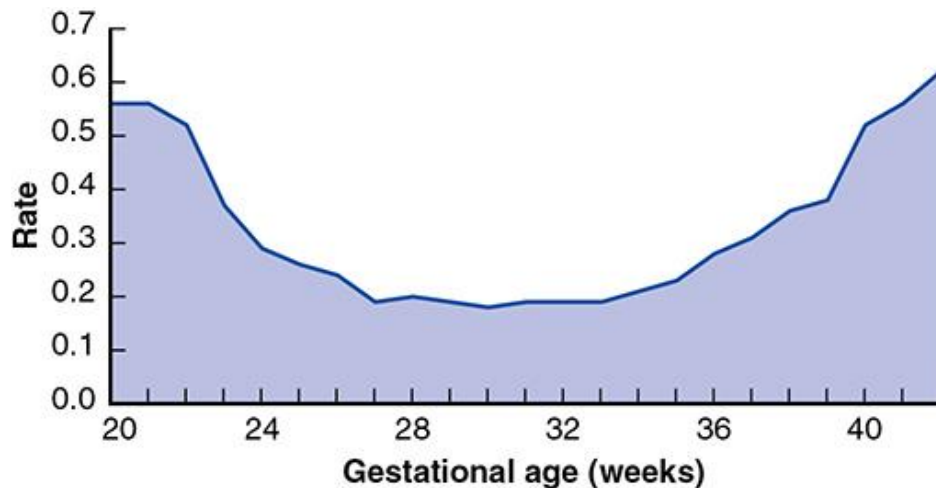


Figure 35-2. Fetal mortality by week is U-shaped: ~0.55 at 20–21 wk, nadir ~0.2 at 27–33 wk, rising to ~0.6 at 42 wk.

3. Etiologies

Stillbirth Collaborative Research Network (SCRN, NICHD)

- **512 stillbirths** (500 women), ≥20 wk, 2006–2008, 59 hospitals in 5 states; population-based.
- Standardized workup: **autopsy, placental histology, maternal/fetal blood and tissue testing, fetal karyotype**.
- **83% died before labor**.
- Causes graded **probable, possible or unknown**.
 - Example: diabetes was *probable* with diabetic embryopathy with lethal anomalies or maternal DKA; *possible* with poor glycemic control plus abnormal fetal growth.
- **Probable or possible cause found in 76%** (similar to Australia).

Table 35-1. Causes of 512 Stillbirths in the Stillbirth Collaborative Research Network Study

Cause	Percent	Examples
Obstetrical complications	29	Placental abruption, multifetal gestation, ruptured membranes at 20–24 weeks
Placental abnormalities	24	Uteroplacental insufficiency, maternal vascular disorders
Fetal malformations	14	Major structural abnormalities and/or genetic abnormalities
Infection	13	Involving the fetus or placenta
Umbilical cord abnormalities	10	Prolapse, stricture, thrombosis
Hypertensive disorders	9	Preeclampsia, chronic hypertension
Medical complications	8	Diabetes, antiphospholipid antibody syndrome
Undetermined	24	Not applicable

Percentages are rounded and total more than 100 percent because some stillbirths had more than one cause. Overall, a probable or possible cause was identified in 76 percent of stillbirths.

- **Leading causes are obstetrical:** abruption, multifetal complications, spontaneous labor or membrane rupture **before viability**; then structural anomalies.
- "Placental abnormalities" is the least clearly defined category (mostly uteroplacental insufficiency).
- The high yield shows the value of **standardized evaluation**, although Cochrane found **no RCT evidence** supporting specific tests.

4. Risk Factors

- **Sociodemographic: non-Hispanic Black women** (more hypertension, diabetes, abruption, PPRM).
- **Maternal:** age at **either extreme**, **obesity**, preeclampsia, diabetes, substance use, **late-term pregnancy**.
- **Multifetal**, especially **monochorionic**.
- **Prior adverse outcome:** stillbirth, preterm birth, FGR.
- **Current FGR** is especially vulnerable (linked to hypertension, diabetes, fetal infection, autoimmune and chronic renal disease).
- **SARS-CoV-2:** data mixed; perhaps higher risk in low-resource countries. At Parkland (**245** infected gravidas) the stillbirth rate was **not** increased.

Table 35-2. Maternal Risk Factors and Estimated Stillbirth Rate

Condition	Estimated Rate (per 1000 births)
All pregnancies	6.4
Low-risk pregnancies	4.0–5.5
Hypertensive disorders	
Chronic hypertension	6–25
Preeclampsia — nonsevere	9–51
Preeclampsia — severe	12–29
Diabetes: insulin requiring	6–35
SLE	40–150
Renal disease	15–200
Thyroid disease	12–20
Cholestasis of pregnancy	12–30
Smoking >10 cigarettes	10–15
Obesity	
BMI 25–29.9 kg/m ²	12–15
BMI >30	13–18
Education (<12 yr vs ≥12 yr)	10–13
Fetal-growth restriction	10–47
Prior fetal-growth restriction	12–30
Late-term pregnancy	14–40
Oligohydramnios	14
Antiphospholipid syndrome	NS
Prior stillbirth	9–20
Multifetal gestation	
Twins	12
Triplets	34
Maternal age	
35–39 yr	11–14
≥40 yr	11–21
Black women (versus white)	12–14

BMI = body mass index; NS = not stated; SLE = systemic lupus erythematosus. The printed table reads "≥49 yr" for the older maternal-age row; this is almost certainly a typo for ≥40 yr (as in the ACOG source table).

Memory hook: The two "hundred-plus" rows are **SLE** and **Renal disease** — "**Stillbirth Rates** soar."

Can risk be predicted early?

- **Consortium on Safe Labor** (206,969 women, 19 hospitals): most stillbirths were at **term**; results **do not support routine antenatal surveillance for any demographic risk factor**.
- **SCRN**: factors known at the start of pregnancy explain only a **small part** of risk. Only **prior stillbirth** or prior loss from **preterm birth or FGR** had meaningful predictive value.
- **Prior stillbirth** → **~5-fold ↑ risk**. Prior preterm birth, FGR, preeclampsia and abruption are strongly linked to later stillbirth.

5. Evaluation of the Stillborn Fetus

Why evaluate: aids maternal coping, eases guilt, allows accurate **recurrence counseling**, may prompt preventive intervention, and identifies **inherited syndromes** for relatives.

- Key tests: **autopsy, chromosomal (genetic) analysis, pathological exam of placenta, cord and membranes.**
- **Placental pathology and fetal autopsy are the most useful** (Page 2017).

Clinical examination

- Measure **fetal weight, head circumference, length, and placental weight.**
- **Photographs:** extremities, palms; frontal and profile views of whole body and face.
- **Fetogram** (full fetal radiograph) may be done.
- If parents decline autopsy, **postnatal MR, radiographs or sonography** become especially important.
- Document findings and relevant ante-/intrapartum events.

Anomaly rates

- With autopsy + chromosomes: **up to 35%** major structural anomalies; **~20%** dysmorphic or skeletal anomalies; **8%** chromosomal abnormalities.
- SCRN: structural anomalies only **14%**.

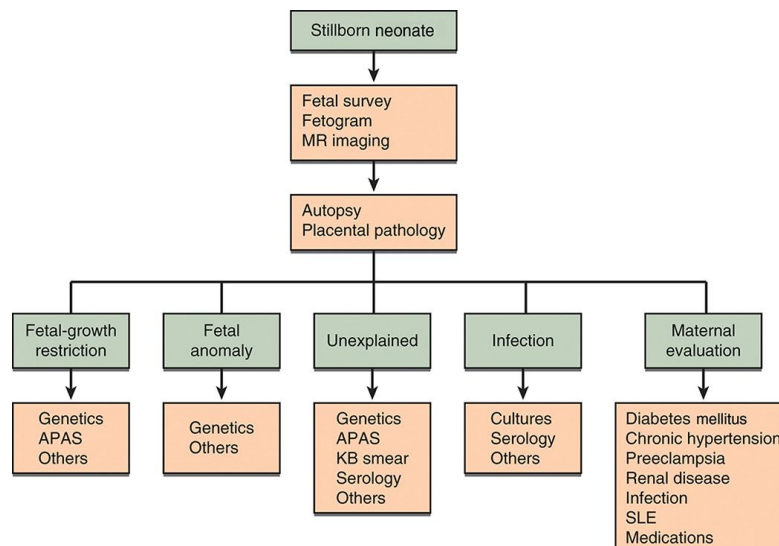


Figure 35-3. Stillbirth workup: fetal survey, fetogram, MR → autopsy and placental pathology → targeted tests by category (FGR: genetics, APAS; anomaly: genetics; unexplained: genetics, APAS, KB, serology; infection: cultures, serology; maternal: diabetes, hypertension, preeclampsia, renal, SLE, medications).

Laboratory evaluation

Genetic testing

- Without dysmorphism, **up to 5%** still have a chromosomal abnormality.
- **Chromosomal microarray analysis (CMA) is now recommended** (replaces routine karyotype): it **does not need dividing cells**, so it works on **macerated tissue** where culture often fails.
- **Exome sequencing** (246 unexplained stillbirths): **8.5%** with normal CMA and no maternal/obstetrical cause were probably due to **mendelian disorders**; a similar number may involve loss-of-function variants in genes essential to fetal survival.
- Requires **parental consent**. Any fetal or placental tissue or amniotic fluid can be sent for CMA. Avoid maternal contamination; **sterile technique, room temperature**.

Specimens if fetal blood (cord or heart) cannot be drawn — send at least one:

- (1) **Placental block 1 × 1 cm** from **below the cord insertion, not formalin-fixed**
- (2) **Umbilical cord segment 1.5 cm** long

- (3) Internal fetal tissue: **costochondral junction, fascia lata or patella**
- Wash in sterile saline; place in **Ringer solution or sterile cytogenetic medium**. ⚠ **Formalin or alcohol kills viable cells.**
- If only conventional karyotype is available and death is recent: **amniocentesis before delivery** gives better cell growth, especially if delivery is not imminent.

Maternal blood tests

Test	Indication
Kleihauer-Betke	Routine (fetomaternal hemorrhage)
Antiphospholipid antibodies / lupus anticoagulant	If indicated
Serum glucose	Exclude overt diabetes
Syphilis serology	If not previously screened, exposure risk, or suggestive maternal/fetal/placental findings
⚠ Inherited thrombophilia panel	Not routinely — most maternal and fetal thrombophilias are not linked to stillbirth (ACOG/SMFM)

Placental examination

- Ideally gross and microscopic exam of **placenta, membranes and cord by a perinatal pathologist** (e.g., cord knots, placental microvascular lesions, infection).

Autopsy

- Encourage a **full autopsy** (SCRN protocol, Pinar 2012).
- Wales (400 deaths): autopsy **changed the cause of death in 13%** and **added new information in 26%**.
- Autopsy changed recurrence estimates and counseling in **25–50%**; placental exam + autopsy altered future management in **45%**.
- **Limited autopsy** still helps: external exam, photographs, radiography, MR, bacterial cultures, selective chromosomal/histological studies.
- Most hospitals do **not** audit stillbirths. Parkland: monthly **stillbirth committee** (obstetricians, MFM, neonatology, genetics, perinatal pathology) assigns cause; parents are then **counseled on cause, recurrence risk and prevention**.

6. Perinatal Care

- Fetal death is psychologically traumatic. **Added stressors:**
 - **>24 h between diagnosis and induction**
 - Not seeing the baby for as long as desired
 - **No keepsakes**
 - Poor communication
- **Seeing and holding the stillborn** helps parental well-being. Few providers are formally trained in bereavement care.
- **Parkland:** dedicated L&D nursing team — time with the neonate, keepsakes, photographs, chaplaincy, bereavement information (matches global consensus).
- After stillbirth or early miscarriage: **↑ depression and PTSD** → monitor closely.
- **Perinatal palliative care** (life-limiting fetal conditions): maximize comfort and quality of life. Components: diagnosis review, **birth plan**, bereavement counseling, neonatal consultation; team gives medical, psychological and spiritual support centered on the parents' values, goals and fears.

7. Prior Stillbirth

- Substantial recurrence risk (Table 35-2); **magnified if the prior fetus was preterm or growth-restricted.**
- SCRNs recurrent stillbirths (43 women): **most died before 28 wk.**

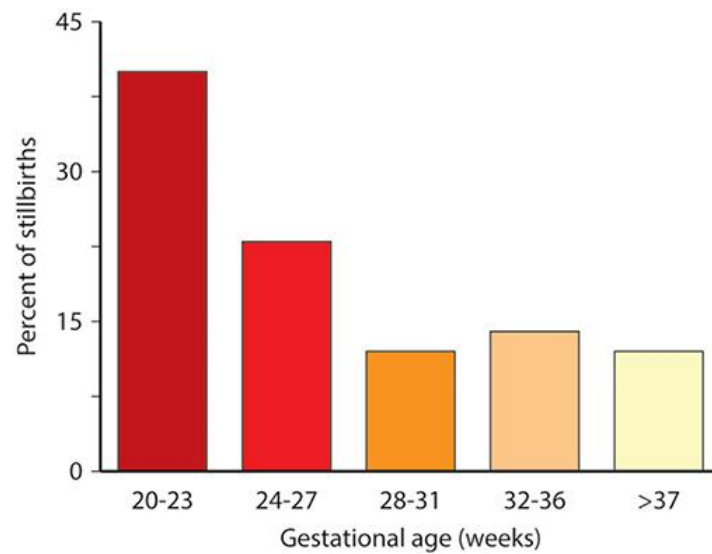


Figure 35-4. Recurrent stillbirths by GA: ~40% at 20–23 wk, ~23% at 24–27 wk, then ~12–14% in each later band.

Table 35-3. Management of Subsequent Pregnancy after Stillbirth

Timing	Management
Preconceptional or initial prenatal visit	Detailed medical and obstetrical history
	Review evaluation of prior stillbirth
	Determination of recurrence risk
	Discuss recurrence of comorbid obstetrical complications
	Smoking cessation
	Preconceptional weight loss in obese women
	Genetic counseling if familial genetic condition exists
	Diabetes screen
	Screen for antiphospholipid antibodies
	Reassurance
First trimester	Dating sonography
	First-trimester screen: PAPP-A, hCG, cfDNA, NT
	Reassurance
Second trimester	Sonographic anatomy survey at 18–20 weeks
	Maternal serum screening (quadruple) <i>or</i> single-marker AFP if first-trimester screening elected
	Possible uterine artery Doppler at 22–24 weeks
	Reassurance
Third trimester	Sonographic screening for FGR starting at 28 weeks
	Kick counts starting at 28 weeks
	Antepartum fetal surveillance starting at 32 weeks <i>or</i> 1–2 weeks earlier than prior stillbirth
	Reassurance
Delivery	Labor induction at 39 weeks
	Delivery before 39 weeks for obstetrical indications

AFP = alpha-fetoprotein; cfDNA = cell-free DNA; hCG = human chorionic gonadotropin; NT = nuchal translucency; PAPP-A = pregnancy-associated plasma protein-A.

- These recommendations rest mainly on **limited or inconsistent evidence** or expert opinion.
- **Few risk factors are modifiable**; treat hypertension or diabetes specifically. Prior placental insufficiency → ↑ later adverse outcome.
- **Almost half of fetal deaths are associated with FGR** → anatomy scan at midpregnancy, then **serial growth scans from 28 wk**. Detecting FGR may cause iatrogenic prematurity but fewer fetal deaths.
- **Weeks (300 women, prior stillbirth only)**: 1 subsequent stillbirth; only **3** abnormal tests before **32 wk**. The GA of the prior loss did **not** predict timing of jeopardy → **start surveillance at ≥32 wk** in otherwise healthy women (ACOG/SMFM agree, noting ↑ iatrogenic preterm delivery).
- **Kick counts**: routinely used but few data for this group.
- **Delivery at 39 wk** unless other indications: minimizes fetal death (maybe more benefit in older women). **Induction is suitable; cesarean only if induction is contraindicated.**

8. Changes in Stillbirth Rates

- US fetal mortality fell until **2011**, then has changed little.
- **"39-week rule"** (no nonmedically indicated delivery <39 wk):
 - 45 states + DC: births <39 wk fell from 2007 to 2013, but **term stillbirth rose** → possible unintended harm.
- **Prospective stillbirth risk** (15 million pregnancies): **0.11 per 1000 at 37 wk → 3.18 per 1000 at 42 wk.**

Method (MacDorman 2006–2012)	Denominator	Finding
Traditional stillbirth rate	Live births + stillbirths at that GA	↑ at 24–27, 34–36 and 38 wk
Prospective stillbirth rate	Women still pregnant at that GA (21–42 wk)	No change

- The discrepancy mainly reflects **fewer preterm and early-term births** (smaller denominators).
- **Little 2015** (early-term 37 0/7–38 6/7 wk, 2005–2011): early-term deliveries fell with no overall change in term stillbirth, but **term singleton stillbirth rose 25% in women with diabetes** — early-term policies were **misapplied to high-risk women.**
- Bottom line: the 39-week rule cut elective early births; indicated early delivery must still be offered to women with complications.

Quick-Recall: Stillbirth Workup and Prevention

Question	Answer
Definition threshold (most US states)	≥20 wk and/or ≥350 g
Epochs	Early <20 wk; intermediate 20–27 wk; late ≥28 wk
Commonest SCRN category	Obstetrical complications (29%) > placental (24%) = undetermined (24%)
Most useful tests	Autopsy + placental pathology
Genetic test of choice	Chromosomal microarray (no dividing cells needed)
Best tissue if no fetal blood	Placental block 1 × 1 cm below cord insertion (unfixed), 1.5-cm cord, or costochondral junction/fascia lata/patella
Transport medium	Sterile saline wash → Ringer or cytogenetic medium; never formalin/alcohol
Maternal blood	Kleihauer-Betke, glucose, ± APLA/lupus anticoagulant, ± syphilis; no routine thrombophilia screen
Highest-risk conditions	SLE (40–150/1000), renal disease (15–200/1000)
Next pregnancy: surveillance start	32 wk or 1–2 wk before prior loss
Next pregnancy: growth scans / kick counts	From 28 wk
Next pregnancy: delivery	Induce at 39 wk

Key Numbers to Memorize

- Baseline stillbirth **6.4/1000**; low-risk **4.0–5.5/1000**; prior stillbirth **9–20/1000** (~5× risk)
- Reporting threshold **20 wk / 350 g**; using ≥500 g would cut the rate **40%**
- SCRN: **512** stillbirths; **83%** antepartum; cause found in **76%**
- Anomalies: up to **35%** structural (autopsy + karyotype), **8%** chromosomal; SCRN **14%**; up to **5%** chromosomal without dysmorphism; exome finds mendelian cause in **8.5%**
- Autopsy changes cause in **13%**, adds data in **26%**; changes counseling in **25–50%**; alters management in **45%**
- Induction delay **>24 h** after diagnosis adds distress

- Recurrent stillbirth mostly **<28 wk**; almost $\frac{1}{2}$ of fetal deaths involve FGR
- Next pregnancy: anatomy **18–20 wk**; uterine artery Doppler **22–24 wk**; growth and kick counts from **28 wk**; testing from **32 wk**; deliver at **39 wk**
- Prospective stillbirth **0.11/1000 at 37 wk** → **3.18/1000 at 42 wk**; diabetic term stillbirth **↑25%** with misapplied 39-wk rule